Defect Report

BUG-0368 — Resource Exhaustion in Voice Pipeline: OOMError: heap exhausted at 741...

1. Defect Summary

Defect ID	BUG-0368
Title	Resource Exhaustion in Voice Pipeline: OOMError: heap exhausted at 741MB/2048MB ...
Severity	Critical
Priority	Low
Status	REOPENED
Component	Voice Pipeline
Application	APP-012 — SecureAuthentication
Reported Date	2023-04-25
Resolved Date	Pending
Assigned To	ENG-024 — Diana Xavier

2. Traceability

Release	REL-003 — 4.3.2
Test Run	RUN-002 — Regression Cycle 2 — Build 4.2.1
Requirement	REQ-0029 — UI Behavior — Biometric Verification
Req Priority	High
Test Case ID	TC-0671
Test Case Title	Ensure search query execution synchronizes correctly across devices wh...
Execution ID	EXEC-00067
Failure ID	FAIL-0017
Failure Type	Resource Exhaustion
Environment	ENV-009 — Low Network Env
Build Version	5.1.0

3. Problem Description

During execution EXEC-00067, test case TC-0671 failed with Resource Exhaustion. The system exhibited resource exhaustion when the target device is running a legacy firmware version.

4. Reproduction Steps

1. Initiate background sync by pressing the 'Search' button.
2. Launch the DigitalWallet application on the test device.
3. Simulate network drop using the network emulator profile 'LTE-Throttled'.
4. Observe the status indicator for up to 26 seconds.
5. Navigate to Settings > Privacy and tap 'Enable Sync'.

SYNTHETIC DATA — No real Samsung confidential information

5. Expected vs Actual Behaviour

Expected	The system should complete payment processing within the defined SLA.
Actual	The system exhibited resource exhaustion when the target device is running a legacy firmware version...

6. Root Cause

Retry logic does not apply exponential back-off; storm of reconnection attempts saturates the gateway.

7. Resolution

Resolution pending investigation.

8. Related Historical Failures

HIST-0016	Resource Exhaustion	5.5.9	2023-05-02	9	Resolved
HIST-0026	Resource Exhaustion	5.5.9	2021-01-21	1	Open
HIST-0045	Resource Exhaustion	5.2.5	2022-12-13	2	Resolved

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

SYNTHETIC DATA — No real Samsung confidential information